

Intelligent Information System

 Knowledge Graphs

 RAG Pipelines

 Databases
Relational / Graph

 Provenance Metadata
PROV-O & logs

 Workflow Logs
Actions and audit trails

Dependency Graph Construction & Artifact Extraction



 Entity Bindings

 Relation Paths

 Evidence Records

 Verification Artifacts

 Provenance Capture

SCAR (Structural Contract for Audit Records)

Identity & Snapshot

 Artifact ID

 Auditable Status

 Graph Snapshot Identifier & digest

Structural Fields

 Bottleneck Status

 Redundancy Status

 Redundancy Group ID

 Downstream Impact Count

 Sink Drop Count

Risk & Impact

 Raw Risk Score
Impact-normalized tie-breaker

 At-risk Terminals Identifier list

Source & Run Metadata

 Candidate Rule

 Protocol Version

 Source Unit

 Replicate

 Candidate-set Digest

Design Requirements

1

Snapshot Identity

2

Candidate Reproducibility

3

Structural Role Preservation

4

Dependency Consequences

5

Policy-independent Consumption

-----Policy-independent Representation Contract-----

Policy Consumers Shared SCAR

 Life-Saving First (LSF)

 Greedy Coverage

 Flat Top-K / Centrality

 Additional Consumers

Illustrative Downstream Maintenance Uses

 Audit Prioritization

 Review Planning

 Maintenance Scheduling

 Governance Support

 Provenance & Traceability